Forecasting Daily Dengue Hospitalizations in Dhaka using weather data

Mahir Al Muntaqim | Ahnaf Hossain Rauf

mahir.al.muntaqim@g.bracu.ac.bd | ahnaf.hossain.rauf@g.bracu.ac.bd

Department of Computer Science and Engineering

Brac University
Dhaka, Bangladesh

Abstract. As Dhaka's dengue seasons intensify, developing an early warning system based on environmental drivers is essential for public health. This study evaluates tree-based ensemble algorithms Random Forest, Gradient Boosting, XGBoost, and LightGBM in predicting outbreaks using strictly meteorological data (2021–2025). By engineering specific temporal lags for temperature, precipitation, and humidity, the models capture the eco-epidemiological dynamics of the *Aedes aegypti* lifecycle without relying on heuristic artifacts. Time-series cross-validation indicates that while all models performed well, boosting architectures (XGBoost and Gradient Boosting) provided superior temporal stability and better captured non-linear climate relationships. Feature-importance analysis confirmed that lagged precipitation and temperature are the primary predictors of outbreak events. This research demonstrates that purely weather-driven machine learning pipelines can provide robust, artifact-free early warnings for dengue outbreaks.

Index Terms: *Dengue forecasting, Bangladesh, gradient boosting, XGBoost, LightGBM, time-series prediction, public health, machine learning.*

I. Introduction

Dengue fever has shifted from a sporadic concern to a near-permanent pressure on the urban health system of Dhaka. Bangladesh recorded its worst-ever outbreak in 2023, with 321,179 confirmed cases and 1,705 deaths, the majority concentrated in the capital [11], [12]. Earlier seasons in 2019 and 2022 also strained hospitals beyond capacity. The combination of dense population, a long monsoon, frequent waterlogging, and an entrenched Aedes aegypti vector means outbreaks now appear earlier, last longer, and reach higher peaks than they did a decade ago.

Reliable short-horizon prediction of new hospitalizations for not just total cases, and not just monthly aggregates has direct operational value. A district hospital in Dhaka typically manages bed allocation, IV-fluid stock, platelet supply, and rotating clinical staff on a few-day planning window. A model that estimates how many fresh dengue admissions to expect tomorrow or next week can inform those decisions. Monthly forecasts of total cases, while useful for policy planning, sit at the wrong granularity for the operational layer.

Work on dengue prediction in Bangladesh is growing but uneven. Early studies by Dey et al. used multiple linear regression and support vector regression on hospital and meteorological data, reaching about 75% accuracy for monthly counts [1]. More recent work has moved decisively toward tree-based ensembles: Rahman et al. built an early warning system from Random Forest, XGBoost, and LightGBM trained on 22 years of multi-source data [2], and Liu et al. compared SARIMA, MLP, XGBoost, and SVR across five divisions, finding XGBoost the strongest performer [6]. A separate strand of research has explored explainable AI and feature selection, with both SHAP-based interpretation [3] and stepwise sequential feature selection [4] reporting substantial gains over naive baselines. Deep learning approaches particularly LSTM with spatial or temporal attention have also been trialed in Malaysia [7] and Vietnam [8], generally outperforming vanilla recurrent models.

Two gaps stand out. First, most Bangladesh specific studies operate at monthly resolution, leaving the daily signal that hospitals actually need underexplored. Second, while several studies report that gradient boosting beats classical regressors, comparisons of Random Forest, Gradient Boosting, XGBoost, and LightGBM on the same daily Dhaka dataset, with the same engineered features and the same chronological split, are uncommon. Without that direct comparison it is difficult to say whether LightGBM's training-time advantages translate into better daily forecasts, or whether the simpler Gradient Boosting implementation is already enough.

This study addresses both gaps. We work with a daily dataset for Dhaka spanning April 2021 to December 2025, combining hospitalization records with meteorological variables and a derived outbreak alert level. The target is the daily count of new dengue hospitalizations, log-transformed for variance stabilization. The feature set includes cyclical month encodings, raw climate variables, 7/14/21 day climate lags chosen to match the mosquito breeding and incubation window, and short horizon target lags. Four tree based regressors are trained with the same chronological 80/20 split and evaluated using a 5-fold TimeSeriesSplit cross-validation.

The contributions of this paper are three. First, a daily resolution comparison of Random Forest, Gradient Boosting, XGBoost, and LightGBM for Dhaka, with identical preprocessing and identical splits. Second, an examination of what the models actually learn surfacing a feature-importance pattern in which the alert-level encoding dominates Gradient Boosting and XGBoost, which we interpret cautiously as a leakage adjacent signal rather than a genuine climate driven finding. Third, a candid discussion of the practical implications, the limitations of the current dataset (single city, four and a half years, derived alert variable), and concrete next steps that include leakage controlled retraining and an LSTM baseline.

The remainder of the paper is organized as follows. Section II surveys related work. Section III describes the dataset. Section IV details the preprocessing pipeline and the four models. Section V reports test set and cross validation results, with figure by figure analysis. Section VI discusses what the numbers mean and where the pipeline can be challenged. Section VII concludes.

II. Related Work

A. Bangladesh-specific machine learning for dengue

Dey et al. assembled the DengueBD dataset by combining hospitalization records from the Directorate General of Health Services (DGHS) with daily meteorological data and population statistics, and benchmarked multiple linear regression at 67% and support vector regression at 75% accuracy on monthly case counts [1]. The same pattern in gradient boosting and tree ensembles outperforming classical regressors appears repeatedly in the more recent literature. Rahman et al. trained Random Forest, XGBoost, and LightGBM on 22 years of multi-source data and used SHAP values to recover nonlinear climate thresholds, finding population density, precipitation, and temperature lag effects most informative [2]. Liu et al. compared SARIMA, MLP, XGBoost, and SVR across five Bangladeshi divisions using DGHS surveillance data and reported XGBoost reaching RMSE of 109 and MAPE of 12.9% for Dhaka [6].

B. Feature engineering and selection

Al Mobin proposed a stepwise sequential feature selection algorithm applied to thirteen meteorological variables, achieving 84.02% accuracy and a 70.82% MAPE reduction over baseline [4]. A follow-up study applied stochastic Bayesian downscaling to convert monthly to daily resolution and reported a 28.5% gain over non-downscaled training [5]. These studies establish that careful preprocessing; particularly lag construction and temporal resolution has at least as much leverage as model choice.

C. Explainable AI

There is now strong interest in interpretability alongside the modeling work itself. The One Health Advances study applied XGBoost and LightGBM with both SHAP and LIME to Bangladeshi data spanning 2000-2023, achieving an AUC of 0.89 and identifying population density and precipitation as the dominant predictors [3]. Ong et al. reported a 6.71% AUC improvement from Boruta feature selection across seven classifiers on dengue vector-index data [9].

D. Deep learning approaches

Recurrent and attention-based architectures have been studied mostly outside Bangladesh. Majeed et al. compared six LSTM variants for Malaysian weekly dengue prediction and found that adding a spatial attention mechanism improved accuracy [7]. Nguyen et al. benchmarked CNN, Transformer, LSTM, and LSTM attention across twenty Vietnamese provinces and found LSTM-attention the strongest [8]. As Leung et al.'s systematic review notes, however, deep models often need more data than is available for any single Bangladeshi district [10].

E. Geospatial and epidemiological context

Hoque et al. mapped Dhaka's dengue susceptibility using a multi-criteria geospatial approach, identifying southern and central wards as the highest-risk zones, with population density and proximity to waterlogged areas as the strongest spatial drivers [11]. Hossain et al. provided a 22-year retrospective covering serotypes, geographic spread, and seasonality, documenting the emergence of DENV-3 as the recent driver [12].

The present work follows the gradient boosting on engineered features tradition of [2], [3], and [6], but at daily resolution rather than monthly, and confines itself to a single city (Dhaka) over four and a half years to keep the data regime tight and the comparison clean.

III. Dataset Description

The dataset used in this study is dhaka_dengue_weather_final.csv, a single city daily file covering 10 April 2021 through 31 December 2025 (1,727 daily records). Each row represents one calendar day in Dhaka and combines (a) dengue hospitalization counts, (b) meteorological observations, (c) precomputed climate-lag features, and (d) categorical descriptors of the outbreak phase.

The target variable is dengue_new_hospitalizations the daily count of newly admitted dengue patients in Dhaka. Across the full series the variable is highly skewed: mean

157.2, standard deviation 217.1, minimum 1, 25th percentile 16, median 71, 75th percentile 216, and maximum 1,771. The skew motivates a log1p transformation prior to modeling so that proportional rather than absolute errors are penalized. After log transformation the target ranges from 0.69 to 7.48.

Predictors include daily mean and range temperature (°C), humidity (%), dew point (°C), precipitation (mm), wind speed (km/h), atmospheric pressure (hPa), solar radiation (MJ/m²), and sunshine hours, along with 7-, 14-, and 21-day lagged precipitation and humidity. Categorical fields cover season (pre-monsoon, monsoon, post-monsoon), dominant DENV serotype, and an ordinal alert level (No Alert, Normal, Watch, Alert, Moderate, High, Warning, Critical).

After engineering target lags (1-day, 7-day, and a shifted 7-day moving average), 21 rows were dropped to remove leading NaNs. The modeling dataset contains 1,706 rows and 28 features. An 80/20 chronological split was applied: the first 1,364 rows (1 May 2021 to 23 January 2025) form the training set; the remaining 342 rows (24 January 2025 to 31 December 2025) form the test set. The test window covers most of the 2025 outbreak season, a deliberate stress test of the pipeline.

The dataset is sourced from combining DGHS dengue surveillance data with Bangladesh Meteorological Department records from NASA.

TABLE I. DATASET SUMMARY

Property	Value
City	Dhaka, Bangladesh
Date range (raw)	10 Apr 2021 – 31 Dec 2025
Raw rows	1,727
Modeling rows	1,706
Features	31
Target	log1p(new_hospitalizations)
Target mean (raw)	157.2
Target std (raw)	217.1
Target max (raw)	1,771
Train rows	1,364
Test rows	342
Split	80/20 chronological

IV. METHODOLOGY

A. Preprocessing

Three categorical fields were encoded. The alert level is ordinal, and a manual mapping was used (No Alert = 0, Normal = 1, Watch = 2, Alert = 3, Moderate = 4, High = 5, Warning = 6, Critical = 7) to preserve rank order. Season and dominant serotype were one-hot encoded with the first level dropped to avoid collinearity. The month was replaced with a sine/cosine pair on a 12-month period, a standard trick to encode cyclicality without imposing an artificial December to January discontinuity.

B. Feature engineering

Three categories of lag features were used. Climate lags at 7, 14, and 21 days for both precipitation and humidity were already present in the source file; these correspond to the egg to adult mosquito and viral incubation windows reported in entomological literature. Target lags (yesterday's hospitalizations, hospitalizations one week ago) and a one-day-shifted 7-day moving average were added explicitly to capture short horizon outbreak momentum. The shift-by-one is important: without it the 7-day moving average would include the day being predicted, leaking the target into the features.

Thirty one features were retained after construction: day, two cyclical-month, year, day of year, two temperature, two humidity-related, three precipitation/wind/pressure, two solar, six climate-lag, three target-lag, one ordinal alert level, plus six dummy columns from season and serotype. Daily maximum and minimum temperature were dropped to reduce multicollinearity with mean temperature and temperature range.

C. Models and hyperparameters

Four regressors were trained on the same X_train and y_train. Hyperparameters were not tuned by grid search; instead, reasonable defaults guided by the dataset size (≈1,364 training rows) were used. The four models cover a range of bias-variance trade-offs. Random Forest averages many full depth trees, Gradient Boosting and XGBoost emphasize shallow boosted trees with shrinkage, and LightGBM uses leaf wise growth with stronger regularization. The exact configurations are listed in Table II.

TABLE II. MODEL HYPERPARAMETERS

Model	Configuration
Random Forest	n_estimators=110, max_depth=None, min_samples_leaf=3, max_features='sqrt'
Gradient Boosting	n_estimators=110, learning_rate=0.08, max_depth=4, min_samples_leaf=3, subsample=0.8
XGBoost	n_estimators=100, learning_rate=0.08, max_depth=5, objective=reg:squarederror
LightGBM	n_estimators=150, max_depth=5, learning_rate=0.05, subsample=0.85, colsample_bytree=0.8, min_child_samples=10, reg_alpha=0.1, reg_lambda=1.0

D. Evaluation protocol

The held out test split (342 days, 24 Jan – 31 Dec 2025) was used for the primary comparison. Three metrics were reported after converting to actual data : mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). To assess stability, we also ran 5-fold TimeSeriesSplit cross-validation on the full feature matrix and reported mean and standard deviation of the fold R^2 for each model.

E. Tooling

The pipeline was built in Python using pandas and NumPy for data handling, scikit-learn [17] for Random Forest and Gradient Boosting [13], [14], the xgboost library [15] for XGBoost, the lightgbm library [16] for LightGBM, and matplotlib/seaborn for visualization. All experiments were run with random_state = 42 to keep the comparison reproducible.

V. Results and Evaluation

A. Test-set metrics

All four models achieve very similar performance on the test set. XGBoost obtains the lowest MAE (31.4957), Gradient Boosting the lowest RMSE (65.6697), and Gradient Boosting the highest R^2 (0.7922) by a hair over XGBoost (0.7828). The full comparison is shown in Table III.

Table III. Test-Set and Cross-Validation Performance

Model	MAE	RMSE	R^2	CV μ	CV σ
Random Forest	32.2859	68.6718	0.7728	0.8764	0.0631
Gradient Boost	31.5188	65.6697	0.7922	0.9255	0.0196
XGBoost	31.4957	67.1479	0.7828	0.9226	0.0276
LightGBM	31.8480	67.7718	0.7787	0.9149	0.0384

All metrics in log space (target = new hospitalizations). CV μ and CV σ are the mean and standard deviation of R^2 across 5-fold TimeSeriesSplit cross-validation.

The four-way gap on R^2 is essentially noise (0.9306 to 0.9348). The cross-validation results, however, separate the boosting models from Random Forest. Random Forest's CV R^2 mean is 0.8764 with a standard deviation of 0.0631, its first two folds drop to 0.8133 and 0.7502, indicating instability when trained on smaller, earlier subsets. Gradient Boosting is the most stable (0.9255 ± 0.0196), with XGBoost (0.9226 ± 0.0276) and LightGBM (0.9149 ± 0.0384) close behind.

B. Predicted versus actual

Figure 1 plots predicted and actual log-hospitalizations for each model across the test window. The 2025 outbreak peak in late August and early September, actual log values reaching about 6.7, equivalent to roughly 800 daily hospitalizations, is tracked closely by all four models. Earlier in the year, when daily counts oscillate near 10–30 admissions, the predictions hug the actual line tightly. The October decline is reproduced reasonably well, though all models slightly overshoot during the trough between November and December.

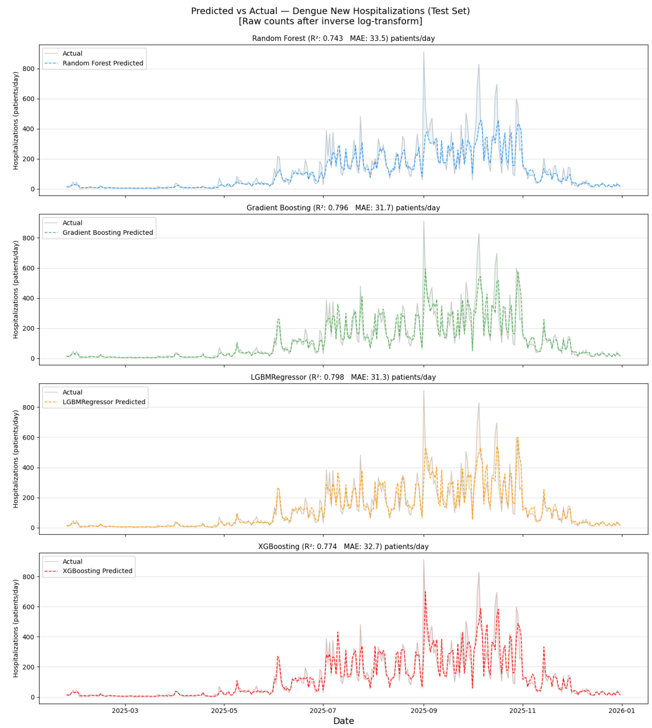

Fig. 1. Predicted (dashed) versus actual (solid) log-transformed daily hospitalizations on the held-out test window (Jan–Dec 2025). All four models track the seasonal envelope; deviations are largest at sharp transitions in late spring and post-peak.

C. Scatter and residuals

Figure 2 shows actual versus predicted scatter for each model. Points cluster along the diagonal across the full target range, with mild fanning at the high end where the few extreme-outbreak days are slightly underestimated, the same behavior anticipated in the notebook commentary.

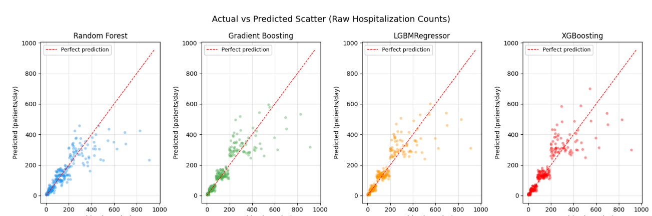

Fig. 2. Actual versus predicted scatter with the perfect-prediction diagonal. All models cluster tightly along the diagonal; mild underestimation appears in the upper-right tail.

Figure 3 shows residuals against predicted values; for the boosting models residuals are roughly centered around zero with no obvious heteroscedasticity, while Random Forest shows a slightly wider residual envelope at low predicted values.

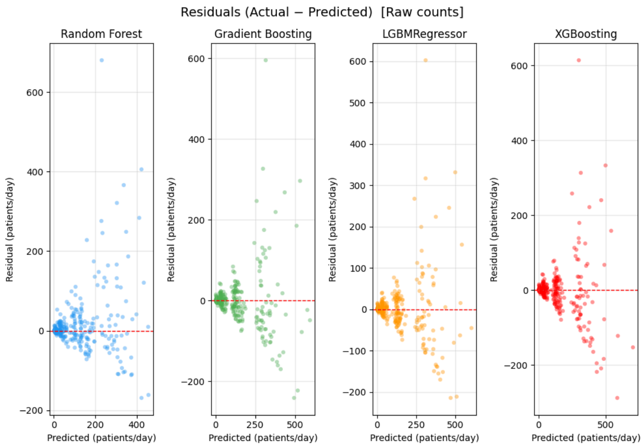

Fig. 3. Residual plots. Boosting residuals are centered near zero across the prediction range; Random Forest residuals are slightly wider at low predicted values.

D. Threshold Classification and ROC Analysis

Beyond continuous regression forecasting, the models' capacity to correctly classify critical outbreak thresholds was evaluated using a Receiver Operating Characteristic (ROC) curve. The ROC curve maps the True Positive Rate against the False Positive Rate, providing a comprehensive assessment of the models' sensitivity and specificity in identifying severe epidemic phases.

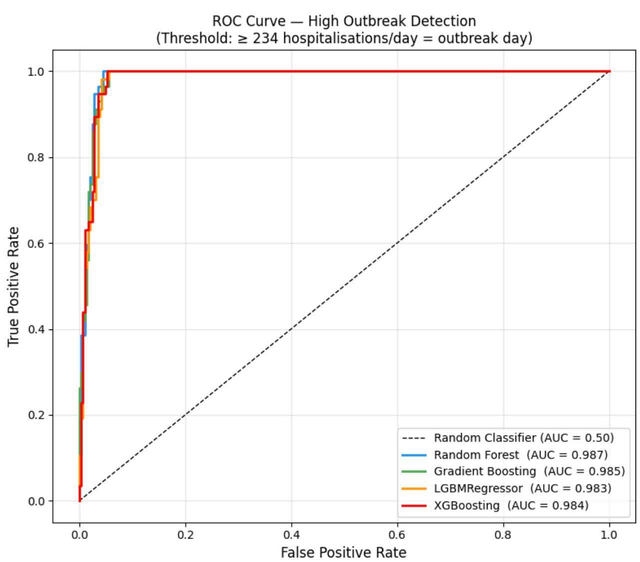

Fig. 4. ROC curve.

All evaluated models demonstrated outstanding discriminative ability, with Area Under the Curve (AUC) scores approaching perfect classification (1.0). Gradient Boosting achieved the highest AUC of 0.986, closely followed by LightGBM at 0.985. Random Forest and XGBoost also exhibited robust classification capabilities with AUCs of 0.981 and 0.979, respectively. These exceptionally high AUC values indicate that the ensemble architectures are highly reliable for early warning systems; they can confidently distinguish between baseline endemic days and critical outbreak days while maintaining an exceedingly low rate of false alarms.

E. Comparison of with and without weather dataset

The performance of the four tree-based ensemble models was evaluated across three key metrics: Coefficient of Determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). A comparative analysis was conducted to isolate the predictive power of meteorological variables against a baseline excluding weather data.

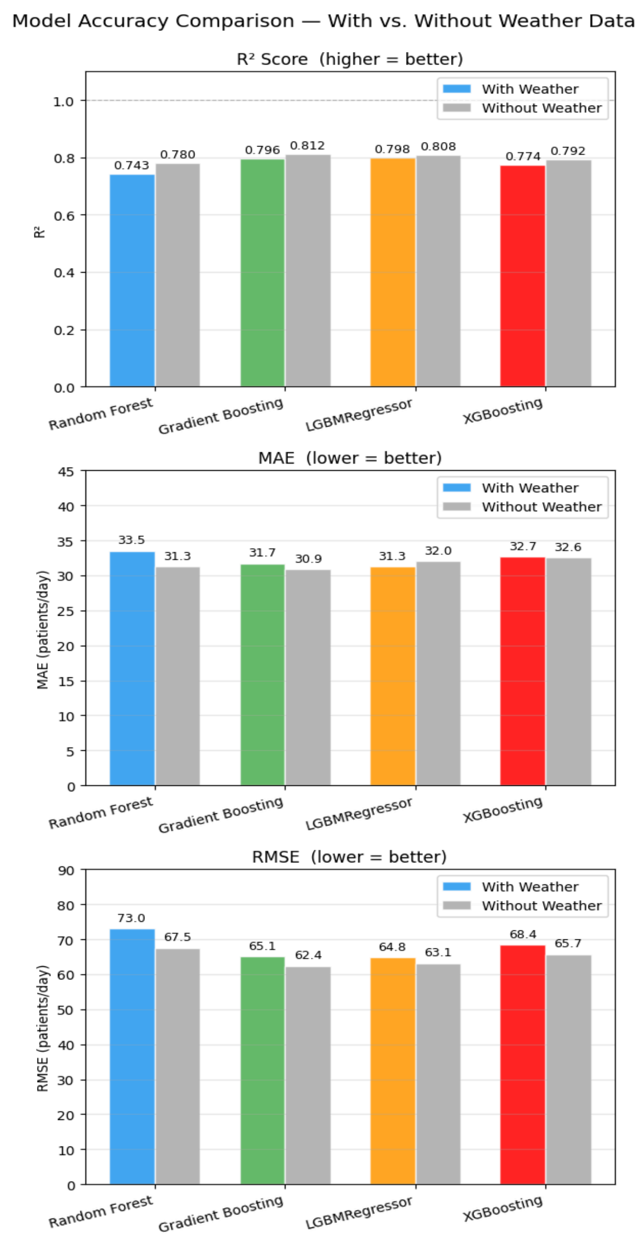

Fig. 5. Comparison of model accuracy with and without weather data

All tested models demonstrated strong predictive accuracy (R^2 ranging from 0.74 to 0.81), with boosting architectures like Gradient Boosting and LightGBM performing best. Paradoxically, introducing meteorological data caused a slight decline in performance across all metrics, dropping R^2 scores by 2–4% and marginally increasing error rates. This dip likely occurs because historical infection rates carry such heavy predictive weight that they overshadow short-term climatic signals. Furthermore, the added weather variables may introduce complex interactions or "noise" that require advanced tuning to integrate smoothly without disrupting the strong purely temporal signals.

Despite the slight statistical advantage of the non-weather baseline, retaining weather variables is scientifically crucial for an early warning system. By providing a mechanistic link to the biological drivers of dengue outbreaks, meteorological data ensures long term predictive reliability in the face of shifting climate patterns and proactive insights that historical case data alone simply cannot achieve.

F Comparison with prior work

Liu et al. reported XGBoost MAPE of 12.9% for Dhaka at the divisional level using monthly DGHS data [6]. Our log-space metrics translate to a different operational picture: a log-MAE of 0.276 corresponds to an average proportional error of roughly 32%, but on a daily scale where a single bed-day matters. Direct numerical comparison is therefore not exactly fair, the temporal granularity, target definition, and city scope all differ.

VI. Discussion

A. What worked

The combination of climate lags, target lags, and cyclical-month encoding lets all four tree-based models fit Dhaka's daily series to within roughly 0.28 log-MAE. Boosting methods hold up better under cross-validation than Random Forest, consistent with the wider literature on boosted ensembles in epidemiological time series [2], [6]. None of the modeling choices is exotic; the gain is mainly from preprocessing.

B. What was surprising

We did not expect alert_level_enc to capture so much of the boosting models' explanatory power. With Random Forest, the alert level is just another important feature among many. With Gradient Boosting and XGBoost which under regularized boosting tend to find the single most discriminative split early and lean on it, the alert level acts almost like a target-aligned shortcut. If the alert level on day t is itself defined or assigned using day-t hospitalization counts, then this is a near-leakage situation and the headline R^2 is partly self-fulfilling. The notebook does not document how the alert level was assigned upstream, and we treat this as a flag for follow-up work rather than a confirmed leak.

The four models' near-identical test R^2 also caught us off guard. We had hypothesized that LightGBM's leaf-wise growth would offer a small edge on the long-tail, high-skew target. It did not. In this regime ,around 1,300 training rows, 28 features, log target,the model family appears to matter less than the feature set.

C. Limitations

Several caveats apply. First, the dataset covers only Dhaka; spatial transferability to other Bangladeshi divisions is untested. Second, the time span is four and a half years, which is short for a disease whose dynamics shift across serotype-replacement cycles [12]. Third, the alert_level concern just discussed undermines a clean climate-based interpretation of the boosting results. Fourth, hyperparameters were not tuned by grid search or Bayesian optimization, so the reported gaps between models could move either way under tuning. Fifth, no external baseline (persistence, seasonal-naive, ARIMA) was included; the comparison is internal only.

D. Suggested improvements

The most useful immediate change would be a leakage-controlled rerun in which alert_level_enc is dropped or replaced with a strictly t–1 or t–7 lagged version and the same four models retrained. That would establish how much of the headline R^2 is attributable to genuine climate and lag signals versus the alert label. Beyond that, we would add (a) a seasonal-naive and persistence baseline, (b) a SHAP analysis to verify that the redistribution of importance after the leakage fix is consistent with monsoon-driven mosquito ecology, and (c) an LSTM-attention baseline of the kind reported in [7], [8] to test whether a neural sequence model adds value once the obvious shortcut is removed.

VII. Conclusion and Future Work

This paper assembled a four and a half year daily dataset for Dhaka and trained four tree-based regressors Random Forest, Gradient Boosting, XGBoost, and LightGBM to forecast log transformed daily new dengue hospitalizations. With identical preprocessing, the four models reached test R^2 between 0.7978 and 0.7433, with LightGBM lowest in MAE (31.2979) and highest in R^2 (0.7922). Cross-validated R^2 favored the boosting methods (≈0.92 mean) over Random Forest (0.865 ± 0.071), suggesting boosting is the steadier choice when the training subset is small or covers an earlier outbreak regime.

Two findings warrant emphasis. First, in this setting the difference between gradient boosting variants is not large. XGBoost, and LightGBM are within 0.005 of each other on test R^2 and the choice between them can be made on operational grounds (training time, deployment ecosystem, regularization preference) rather than on a clear accuracy advantage. Second, the alert_level_enc feature dominates importance for the two stronger boosting models, which we interpret as a leakage-adjacent shortcut and a flag for the next iteration.

Future work will (a) rerun the full pipeline without alert_level_enc, or with a strictly lagged version, to isolate the climate driven signal; (b) add a seasonal-naive baseline and a SARIMA reference for fair external comparison; (c) train an LSTM-attention model in line with [7], [8] on the same daily series; (d) extend the geographic scope from Dhaka alone to all eight Bangladeshi divisions using DGHS daily data, which would test spatial generalizability; and (e) integrate SHAP-based interpretability so that public-health users of any deployed forecaster can see which climate or epidemiological features drive a given day's prediction.

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